

## Test Certificate

|                           |                                                         |
|---------------------------|---------------------------------------------------------|
| <b>Product Code:</b>      | BMMC1DLC144BK                                           |
| <b>Description:</b>       | Up to 144f, 12 position Stranded loose tube ADSS, AR/PE |
| <b>Cable #:</b>           | 14551/1                                                 |
| <b>Date Cert created:</b> | 24 <sup>th</sup> January 2022                           |
| <b>Design Number:</b>     | 39-16                                                   |
| <b>Product Family:</b>    | BMM*LCxxPx <sup>+++</sup> XX                            |

| TEST PARAMETER / METHOD                            | AFL ACCEPTANCE CRITERIA                                                                                                                                                                                                                                                                                                                                                                                                                                          | ACHIEVED                                                                                                                                              | RESULT              |
|----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| <b>Tensile Strength</b><br>IEC 60794-1-21 (E1)     | <b>13.5kN</b><br>30min, < 0.6% fibre strain, No att. Increase after test, < 0.1dB during test, @ 1550nm                                                                                                                                                                                                                                                                                                                                                          | <b>13.5kN</b><br><0.0% fs<br>$\Delta\alpha \leq 0.05\text{dB}$<br>No att. Increase during test                                                        | <b>Pass</b>         |
| <b>Bend under tension</b><br>IEC 60794-1-21 (E18A) | $r_b = 10 \times \text{OD}$<br>360°, 1 turn, 1 min<br>No fibre break, no damage to sheath or core structure and<br>No attn. change throughout test                                                                                                                                                                                                                                                                                                               | $r_b = 10 \times \text{OD}_{\text{measured}}$<br>360°, 1 turn, 1 min<br>No breaks or damage and no attenuation change                                 | <b>Pass</b>         |
| <b>Crush (short term)</b><br>IEC 60794-1-21 (E3)   | <b>2kN/100mm</b><br>Duration: 10min<br>3 position<br>No damage to sheath or core structure and No att. Increase after test                                                                                                                                                                                                                                                                                                                                       | <b>&gt;2kN/100mm</b><br>No damage and no attenuation increase                                                                                         | <b>Pass</b>         |
| <b>Crush (long term)</b><br>IEC 60794-1-21 (E3)    | <b>2kN/100mm</b><br>Duration: 120min<br>3 positions<br>No damage to sheath or core structure and No att. Increase during or after test                                                                                                                                                                                                                                                                                                                           | <b>&gt;2kN/100mm</b><br>No damage and no attenuation increase                                                                                         | <b>Pass</b>         |
| <b>Impact</b><br>IEC 60794-1-21 (E4)               | <b>Weight: 1kg</b><br><b>Height: 1.0m</b> ,<br>300mm rad anvil,<br>3 impacts,<br>After 5 mins No fibre break, no damage to sheath or core structure and<br>No attn. change throughout test                                                                                                                                                                                                                                                                       | No breaks or damage and no attenuation change                                                                                                         | <b>Pass</b>         |
| <b>Bend</b><br>IEC 60794-1-21 (E11)                | $r_b = 10 \times \text{OD}$<br>360° loop, 1 turn, 1 cycle<br>No att. Change throughout test                                                                                                                                                                                                                                                                                                                                                                      | $r_b = 10 \times \text{OD}$<br>360° loop, 5 turns, 1 cycle<br>No att. Change throughout test                                                          | <b>Pass</b>         |
| <b>Torsion</b><br>IEC 60794-1-21-(E7)              | Sample length: 1 m<br>Rotation: a) 180° clockwise, b) return to starting position, c) 180° anticlockwise, d) return to starting position. Four movements = one cycle. Complete each cycle within 1 min max. for total 10 cycles (a to d)<br><br>During the final tenth cycle at a), c) and after completion (no rotation) check transmitting fibres. No fibre breaks, no damage to the sheath or to the core structure and no attenuation change throughout test | No breaks or damage and no attenuation change                                                                                                         | <b>Pass</b>         |
| <b>Temperature Cycle</b><br>IEC 60794-1-22 (F1)    | Temperature Range: <b>-40°C to +70°C</b><br>$\Delta\alpha_{\text{max}} \leq 0.15\text{dB}$<br>$\Delta\alpha_{\text{avg}} < 0.05\text{dB}$<br>@ 1550nm                                                                                                                                                                                                                                                                                                            | Temperature Range: <b>-40°C to +70°C</b><br>$\Delta\alpha_{\text{max}} \leq 0.15\text{dB}$<br>$\Delta\alpha_{\text{avg}} < 0.05\text{dB}$<br>@ 1550nm | <b>Pass</b>         |
| <b>Water Penetration</b><br>IEC 60794-1-22 (F5B)   | Sample length: 3m<br>Water Height: 1m<br>No water after 24 hours                                                                                                                                                                                                                                                                                                                                                                                                 | No Water detected                                                                                                                                     | <b>Pass</b>         |
| <b>Tested By:</b>                                  | Attila N & Karlo L                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Verified By:</b>                                                                                                                                   | Shashini Subasinghe |
| <b>Approved By:</b>                                | Jim Boukouvalas                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                       |                     |